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12 May 1975
SUPERCEDING
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10 February 1975

MILITARY SPECIFICATION

THEOPHYLLINE AND GLYCERYL GUAIACOLATE CAPSULES.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Theophylline and Glyceryl Guaiacolate Capsules.

1.2 Classification. The capsules covered by this specification shall be the following sizes (See 6.2):

Size 1 - Individually Sealed 100s
Size 2 - Bottle of 1000s

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-C-186 Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals. (See 6.3)

STANDARDS

Federal

FED-STD-00285 Capsules (for Medicinal Purposes).

FSC 6505

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Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN PHARMACEUTICAL ASSOCIATION PUBLICATION

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE, FOOD AND DRUG ADMINISTRATION

Federal, Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U.S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be Theophylline and Glyceryl Guaiacolate Capsules, and shall be in accordance with all applicable requirements of FED-STD-00285, and as specified herein:

S2 Classification. Shall be Type II.

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S5.4 The following additional requirements and tests are added to this paragraph:

Each capsule shall contain 150 mg of anhydrous theophylline and 90 mg of glyceryl guaiacolate, within the designated assay limits for the capsule. The capsule shall contain the active ingredients in a suspension of glycerin, polyethylene glycol 400, and polyethylene glycol 6000.

The capsule shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the required amounts of anhydrous theophylline and glyceryl guaiacolate, when prepared as specified in 4.3.1 and assayed as specified in 4.3.1.1 for glyceryl guaiacolate and 4.3.1.2 for the anhydrous theophylline.

Identity. The active ingredients in the finished capsule shall comply with the following identity tests:

Glyceryl Guaiacolate. The assay solution prepared as directed in 4.3.1.1 for glyceryl guaiacolate, shall exhibit a maximum absorbance at 276 ± 2 mu.

Theophylline, anhydrous. The assay solution prepared as directed in 4.3.1.2 for anhydrous theophylline, shall exhibit a maximum absorbance at 274 ± 2 mu.

pH. A solution of three (3) capsules per 100 ml of purified water shall have a pH of 4.20 to 6.40, at 25° C., when determined potentiometrically using the U.S.P. method.

S6.1 Ingredients. The following additional requirements are added to this paragraph:

The anhydrous theophylline used in the manufacture of the finished capsules shall be in accordance with the tests, standards, and requirements of the USP and, in addition, shall comply with the following:

Particle size. The particle size of the anhydrous theophylline shall be such that a minimum of 98 percent shall pass through a No. 200 U.S. Standard Sieve.

All other ingredients used in the manufacture of the finished capsules shall comply with S5.2.

S6.2.15 Acid solubility. In the second subparagraph, Line 2, delete "5 minutes" and substitute "15 minutes".

S6.7.1 Classification of defects. Inspection for Defect #105 (Capsule, empty), Table V. Soft Capsules, shall be performed by weighing 125 capsules, cutting them open, and determining the weight of fill.

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3.2 Potency Period. For size 1 the potency period (Expiration Dating Period) shall be 24 months and for size 2 the potency period shall be 48 months.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 4 years after delivery of the supplies to which such records relate.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Sampling. The classification of defects as shown in FED-STD-00285 shall be applicable.

4.3 Tests.

4.3.1 Sample preparation. Cut ten (10) capsules in half and quantitatively transfer to a 500-ml volumetric flask. Add approximately 350 ml of purified water and shake until the contents of the capsules have dissolved. Add purified water to volume and mix well. Pipet 15.0 ml of the solution into a 125-ml separatory funnel, add about 15 ml of 1N Sodium Hydroxide solution and mix by swirling. Extract with three 25-ml portions and one 20-ml portion of Chloroform. Drain each chloroform layer through a pledget of cotton in the stem of the separatory funnel into a 100-ml volumetric flask.

Save the chloroform layer for the determination of glyceryl guaiacolate and immediately add 3 ml of concentrated hydrochloric acid to the aqueous layer, mix by swirling and save this layer for the determination of theophylline.

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4.3.1.1 Glyceryl Guaiacolate assay. Using the chloroform extract, above, make to volume with chloroform and mix well. Pipet 15.0 ml into another 100-ml volumetric flask, make to volume with chloroform and mix well. Quantitatively and stepwise, prepare a solution of N.F. Glyceryl Guaiacolate Reference Standard containing approximately 40 mcg per ml in chloroform. Concomitantly determine the absorbances of the sample and standard solutions at the maxima at about 276 and 310 mu in 1-cm cells with a suitable spectrophotometer using chloroform as the blank. Calculate the amount of glyceryl guaiacolate as follows:

$$\text{mg per capsule} = \frac{(Au^{276} - Au^{310}) \times C \times 100 \times 500 \times 100}{(As^{276} - As^{310}) \times 15 \times 10 \times 15 \times 1000}$$

where Au is the absorbance of the sample at the designated wavelengths
As is the absorbance of the standard at the designated wavelengths
C is the concentration of the standard in mcg per ml.

$$\text{\% of required amound} = \frac{\text{mg per capsule}}{90} \times 100$$

4.3.1.2 Theophylline, anhydrous, assay. Use the acidified aqueous extract. Remove the cotton from the stem of the separatory funnel and extract the solution with 50 ml of Diethyl Ether, U.S.P. Drain the aqueous (bottom) phase into a second 125-ml separatory funnel containing 25 ml of ether. Shake the second funnel for about 30 seconds and allow the layers to separate. Drain the aqueous layer into a 200-ml volumetric flask. Drain the ether layer in the second separatory funnel into the first separatory funnel; wash the second funnel with three 3-ml portions of ether and add the washings to the first separatory funnel. Extract with two, 25-ml portions of 0.1N hydrochloric acid. Allow the layers to separate and drain the aqueous layer into the same 200-ml volumetric flask. Add purified water to volume to the volumetric flask and mix. Pipet 5.0 ml of this solution into a 200-ml volumetric flask.

Quantitatively and stepwise prepare a solution of U.S.P. Theophylline Reference Standard containing approximately 250 mcg of anhydrous theophylline per ml. Pipet 5.0 ml of this solution into a 200-ml volumetric flask.

Concomitantly treat the sample and standard solutions as follows:

Add 25 ml 1N Sodium Hydroxide and dilute to volume with purified water. Mix and immediately determine the absorbances at the maxima at about 274 and 320 mu in 1 cm cells with a

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suitable spectrophotometer using purified water as a blank. Calculate the amount of anhydrous theophylline as follows:

mg anhydrous Theophylline per capsule =

$$\frac{(Au^{274} - Au^{320}) \times C \times 500 \times 200}{(As^{274} - As^{320}) \times 10 \times 1000 \times 15}$$

where Au is the absorbance of the sample at the designated wavelength
As is the absorbance of the standard at the designated wavelength
C is the concentration of the standard in mcg per ml

$$\% \text{ of required amount} = \frac{\text{mg per capsule} \times 100}{150}$$

4.4 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table I. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table I. Examination of preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

NOTE: The above examination is only applicable to the unit, intermediate and exterior containers.

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5. PREPARATION FOR DELIVERY

5.1 Preparation for Delivery shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be Level A or C as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Immediate container.

5.2.1.1.1 Size 1. Capsules shall be individually sealed in an immediate container which shall be a commercially available blister or strip package considered acceptable to the Food and Drug Administration (FDA) for the product contained therein. The compatibility shall be such that neither the contents nor the container are altered in any way by each other.

5.2.1.1.2 Size 2. Shall comply with the following: Shall be a commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.1.2 Unit package. At the option of the contractor, a unit package may be furnished for size 2 capsules. If a unit package is furnished, each immediate container (bottle) shall be packaged as specified in 5.2.5 of PPP-C-186. A unit package shall be furnished for size 1 capsules. One hundred individually blister or strip packaged capsules shall be packaged in a box as specified in 5.2.5 of PPP-C-186.

5.2.1.3 Intermediate package. Ten unit packages containing size 1 capsules shall be packaged in an intermediate package as specified in 5.2.8 of PPP-C-186. Twelve immediate containers or unit packages, as applicable, containing size 2 capsules shall be packaged in an intermediate package as specified in 5.2.9 of PPP-C-186.

5.2.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.3 Packing. Packing shall be level A, B, or C as specified (see 6.1).

5.3.1 Level A. Ten intermediate packages containing size 1 capsules or one intermediate package containing size 2 capsules shall be packed in an exterior container as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Ten intermediate packages containing size 1 capsules shall be packed in an exterior container as specified in 5.4.3.2 of PPP-C-186. Unless otherwise specified, no overpacking will be required for size 2 capsules. The intermediate package shall serve as the exterior container.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and shall include the information required below.

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the expiration date.

5.4.1.2 Unit package. Each unit package shall bear the following:

(a) labeling information in accordance with commercial practice.

(b) the national stock number.

NOTE: The National Stock Number shall not be preceded by any prefix.

Strip labeling may be used.

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The label adhesion test as specified in PPP-C-186 shall not apply.

If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type I Shelf life markings as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86°F.)".

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86°F.)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) the National Stock No.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following items:

National Stock Number	Item Identification
6505-00-118-2496	THEOPHYLLINE AND GLYCERYL GUAIACOLATE CAPSULES, INDIVIDUALLY SEALED, 100s
6505-00-764-3366	THEOPHYLLINE AND GLYCERYL GUAIACOLATE CAPSULES, 1000s

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superceded by a revision.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1378
cs

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-T-37144 THEOPHYLLINE AND GLYCERYL GUAIACOLATE CAPSULES

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)

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